WORKED SOLUTION

STEP III 2002, Question 8

Pure • Ptolemy's Theorem

STEP III 2002, Question 8

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/02-s3-q8

The key idea

The difference of two points on the unit circle factorises beautifully if we pull out the "average" angle: $e^{i\alpha} - e^{i\beta} = e^{i(\alpha+\beta)/2}(e^{i(\alpha-\beta)/2} - e^{-i(\alpha-\beta)/2})$, and the bracket is 2i times a sine. That gives both the modulus and the argument of every chord at once. For the last part, two complex numbers with the same argument point in the same direction, so their moduli simply add; all we then need is an algebraic identity linking the three products.

Solution

The argument of $u_1 - u_2$

Write $u_k = e^{i\theta_k}$. Taking out a factor $e^{i(\theta_1+\theta_2)/2}$,

$$\begin{aligned} u_1 - u_2 &= e^{i(\theta_1+\theta_2)/2}\left(e^{i(\theta_1-\theta_2)/2} - e^{-i(\theta_1-\theta_2)/2}\right) \\ &= e^{i(\theta_1+\theta_2)/2} \cdot 2i \sin \tfrac{1}{2}(\theta_1 - \theta_2) \\ &= 2\sin \tfrac{1}{2}(\theta_2 - \theta_1) \cdot (-i)\, e^{i(\theta_1+\theta_2)/2} \\ &= 2\sin \tfrac{1}{2}(\theta_2 - \theta_1)\, e^{i(\theta_1+\theta_2-\pi)/2}, \end{aligned}$$

using $-i = e^{-i\pi/2}$ in the last line.

Ask yourself Why can we read the argument straight off this, and why do we need the sine in the form $\sin \frac{1}{2}(\theta_2 - \theta_1)$ rather than $\sin \frac{1}{2}(\theta_1 - \theta_2)$?

A number of the form $re^{i\phi}$ has argument ϕ (up to multiples of 2π) only when $r > 0$. Here $0 < \theta_2 - \theta_1 < 2\pi$, so $0 < \frac{1}{2}(\theta_2 - \theta_1) < \pi$ and the sine is strictly positive. Written the other way round the sine would be negative, and the argument would be out by π.

So $|u_1 - u_2| = 2\sin \frac{1}{2}(\theta_2 - \theta_1)$ and $\frac{1}{2}(\theta_1 + \theta_2 - \pi)$ is an argument of $u_1 - u_2$. Taking

arg to be the principal value in $(-\pi, \pi]$: since $-\pi < \theta_1 < \theta_2 < \pi$,

$$-\tfrac{3\pi}{2} < \tfrac{1}{2}(\theta_1 + \theta_2 - \pi) < \tfrac{\pi}{2}.$$

If this value is greater than $-\pi$ it is already the principal argument ($n = 0$); otherwise it lies in $(-\frac{3\pi}{2}, -\pi]$ and adding 2π brings it into range ($n = 1$). Hence

$$\arg(u_1 - u_2) = \tfrac{1}{2}(\theta_1 + \theta_2 - \pi) + 2n\pi, \qquad n = 0 \text{ or } 1,$$

as required.

The two products have the same argument

Nothing above used anything about u_1, u_2 except that $\theta_1 < \theta_2$ and both lie in $(-\pi, \pi)$. So for any $j < k$,

$$\begin{aligned} u_j - u_k &= 2\sin\tfrac{1}{2}(\theta_k - \theta_j)\, \mathrm{e}^{\mathrm{i}(\theta_j + \theta_k - \pi)/2}, \\ u_k - u_j = -(u_j - u_k) &= 2\sin\tfrac{1}{2}(\theta_k - \theta_j)\, \mathrm{e}^{\mathrm{i}(\theta_j + \theta_k + \pi)/2}, \end{aligned}$$

using $-1 = \mathrm{e}^{\mathrm{i}\pi}$, and in each case the real factor in front is positive. Write $s_{jk} = \sin\frac{1}{2}(\theta_k - \theta_j) > 0$ for $j < k$, and $\Sigma = \theta_1 + \theta_2 + \theta_3 + \theta_4$. Multiplying (arguments add),

$$\begin{aligned} (u_1 - u_2)(u_4 - u_3) &= 4s_{12}s_{34}\, \mathrm{e}^{\mathrm{i}[(\theta_1 + \theta_2 - \pi) + (\theta_3 + \theta_4 + \pi)]/2} = 4s_{12}s_{34}\, \mathrm{e}^{\mathrm{i}\Sigma/2}, \\ (u_1 - u_4)(u_3 - u_2) &= 4s_{14}s_{23}\, \mathrm{e}^{\mathrm{i}[(\theta_1 + \theta_4 - \pi) + (\theta_2 + \theta_3 + \pi)]/2} = 4s_{14}s_{23}\, \mathrm{e}^{\mathrm{i}\Sigma/2}. \end{aligned}$$

Both are positive multiples of $\mathrm{e}^{\mathrm{i}\Sigma/2}$, so each has argument $\frac{1}{2}\Sigma$ up to a multiple of 2π. Their principal arguments can therefore differ only by a multiple of 2π:

$$\arg((u_1 - u_2)(u_4 - u_3)) = \arg((u_1 - u_4)(u_3 - u_2)) + 2n\pi$$

for some integer n. (In fact $n = 0$, since two principal arguments that differ by a multiple of 2π must be equal, but the weaker statement is all that was asked.)

The final part

Call the three products

$$A = (u_1 - u_2)(u_4 - u_3), \qquad B = (u_1 - u_4)(u_3 - u_2), \qquad C = (u_1 - u_3)(u_4 - u_2).$$

Expanding,

$$\begin{aligned} A + B &= (u_1u_4 - u_1u_3 - u_2u_4 + u_2u_3) + (u_1u_3 - u_1u_2 - u_3u_4 + u_2u_4) \\ &= u_1u_4 + u_2u_3 - u_1u_2 - u_3u_4, \\ C &= u_1u_4 - u_1u_2 - u_3u_4 + u_2u_3, \end{aligned}$$

so $A + B = C$. This is pure algebra and holds for any four complex numbers.

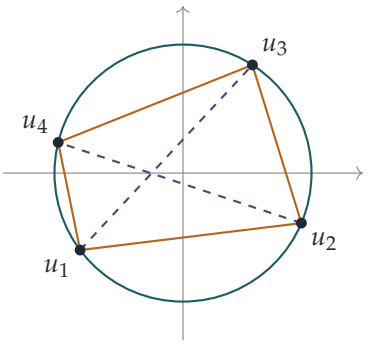
Now use the previous part: $A = a\,\mathrm{e}^{\mathrm{i}\Sigma/2}$ and $B = b\,\mathrm{e}^{\mathrm{i}\Sigma/2}$ with $a = 4s_{12}s_{34} > 0$ and $b = 4s_{14}s_{23} > 0$. Then $|A| = a$, $|B| = b$, and

$$|C| = |A + B| = |(a + b)\,\mathrm{e}^{\mathrm{i}\Sigma/2}| = a + b = |A| + |B|,$$

because $a+b>0$ and $|\mathrm{e}^{\mathrm{i}\Sigma/2}|=1$. That is,

$$|(u_1-u_2)(u_4-u_3)|+|(u_1-u_4)(u_3-u_2)|=|(u_1-u_3)(u_4-u_2)|.$$

The point is that the triangle inequality $|A+B|\le|A|+|B|$ becomes an equality exactly when A and B point the same way, and that is what the middle part established.



Geometrically, $|u_j-u_k|$ is the length of a chord, and because the arguments increase, $u_1u_2u_3u_4$ is a quadrilateral inscribed in the circle with its vertices in order. The identity says that the products of opposite sides add up to the product of the diagonals (dashed): this is Ptolemy's theorem.

Checks

- A square: $\theta_k=-\frac{3\pi}{4},-\frac{\pi}{4},\frac{\pi}{4},\frac{3\pi}{4}$. Each side is $\sqrt{2}$ and each diagonal is 2, and indeed $2+2=4$.
- With $\theta=(-2.5,-0.4,1.0,2.9)$ we get $|A|\approx 2.8223$, $|B|\approx 1.1013$ and $|C|\approx 3.9236=|A|+|B|$.
- Over 2×10^5 random ordered choices of the θ_k, the formula for $\arg(u_1-u_2)$ held with $n=0$ or 1 every time (both values occur), and the identity held to within 2×10^{-15}.

Common slips

- Writing $u_1-u_2=2\mathrm{i}\sin\frac{1}{2}(\theta_1-\theta_2)\,\mathrm{e}^{\mathrm{i}(\theta_1+\theta_2)/2}$ and reading off the argument as $\frac{1}{2}(\theta_1+\theta_2+\pi)$, forgetting that the sine is negative.
- Treating u_4-u_3 as if it had the same argument as u_3-u_4: reversing a difference adds π to the argument.
- Asserting $|A|+|B|=|A+B|$ without saying why; it needs A and B to have the same argument.

Beyond STEP

Ptolemy's theorem has a converse, and a more general form: for any four points in the plane, $|A| + |B| \geq |C|$ (just the triangle inequality applied to $A + B = C$), with equality exactly when the points lie in order on a circle or a line. Behind this lies the *cross-ratio* $\frac{(u_1-u_2)(u_3-u_4)}{(u_1-u_4)(u_3-u_2)}$, which is real precisely when four points are concyclic and is preserved by Möbius transformations, a central object in first-year complex analysis and geometry. Needham's *Visual Complex Analysis* develops these ideas with wonderful pictures.

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